Building Science

Lecture 4: Heat transfer principles

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Lecture outline

- Physics of Heat
- 1st and 2nd Law of Thermodynamics
- Phase changes

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Building Science

Course content

1. General introduction to Building Science
2. Climate impacts on Building orientation
3. Energy efficiency and Passive control strategies
4. Thermal performance – heat transfer principles
5. Thermal comfort
6. Thermal performance – Insulation and thermal bridges

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Physics of Heat

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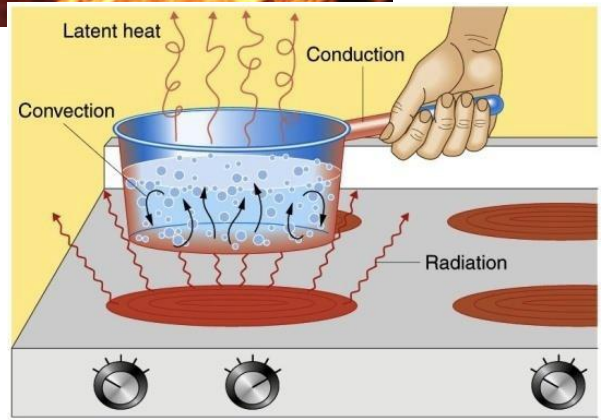
Physics of Heat

What is ‘Heat’?

✓A form of energy, which is transferred as a result of temperature difference

✓Heat is one form of Energy:

- ✓The ability or capability for doing work
- ✓Measured in Joules (J)



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Physics of Heat

Heat is just one form of *Energy*

- ✓ Thermal or heat energy:
  - ✓ To heat up a cup of tea by $\Delta T = T_2 - T_1$

$$Q = c * m * \Delta T$$

- ✓ Q – thermal energy
- ✓ c – thermal heat capacity
- ✓ m – mass (kg)
- ✓ T - absolute temperature

- ✓ Other forms of energy?
  - ✓ Mechanical energy
  - ✓ Electrical energy
  - ✓ Chemical energy
  - ✓ Nuclear energy

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Physics of Heat

What is it?

TO \ FROM	ELECTRO-MAGNETIC	CHEMICAL	NUCLEAR	THERMAL	KINETIC	ELECTRICAL
ELECTRO-MAGNETIC		CHEMILUMINES-CENCE	NUCLEAR BOMBS	THERMAL RADIATION	ACCELERATING CHARGES	ELECTRO-MAGNETIC RADIATION
CHEMICAL	PHOTO-SYNTHESIS	CHEMICAL PROCESSING		BOILING	DISSOCIATION BY RADIOLYSIS	ELECTROLYSIS
NUCLEAR	GAMMA-NEUTRON REACTIONS					
THERMAL	SOLAR ABSORPTION	COMBUSTION	FISSION / FUSION	HEAT EXCHANGE	FRICTION	RESISTANCE HEATING
KINETIC	RADIOMETERS	METABOLISM	RADIOACTIVITY / NUCLEAR BOMBS	THERMAL EXPANSION / INTERNAL COMBUSTION	GEARS	ELECTRIC MOTORS
ELECTRICAL	SOLAR CELLS	FUEL CELLS / BATTERIES	NUCLEAR BATTERIES	THERMO-ELECTRICITY	ELECTRICITY GENERATORS	

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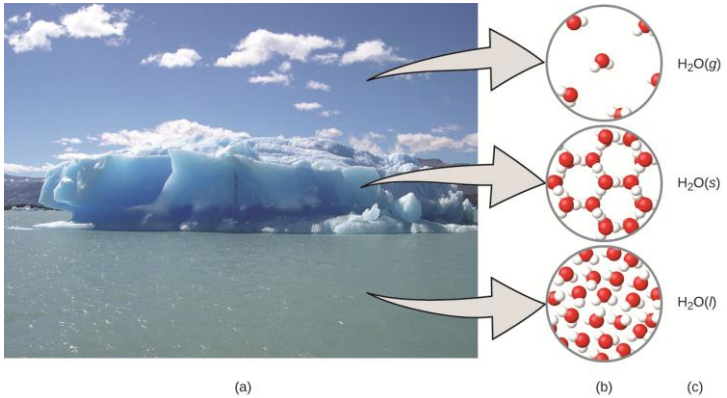
Physics of Heat

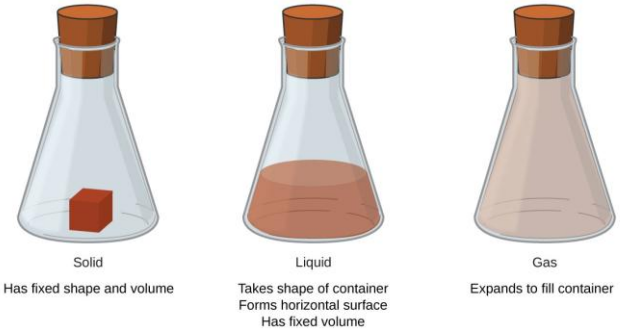
$$Q = c * m * \Delta T$$

What is special about heat?

✓Thermal or heat energy:

- ✓To heat up a cup of tea by $\Delta T = T_2 - T_1$
- ✓What is heat physically?





Heat is the microscopic movement of atoms or molecules

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Physics of Heat

$$Q = c * m * \Delta T$$

What is special about heat?

✓Thermal or heat energy:

- ✓To heat up a cup of tea by $\Delta T = T_2 - T_1$
- ✓Can you feel thermal energy?
  - ✓ No! can only feel TEMPERATURE
    - ✓ Hot = high temperature
    - ✓ Cold = low temperature

Note! There is no human sense for absolute temperature. We can only compare temperatures

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Physics of Heat

$$Q = c * m * \Delta T$$

What is 'Heat'?

✓Sensible Heat

- The quantity of heat needed to *change a material from one phase of a unit mass completely to another phase*.
- (e.g. Ice to water, water to steam)
- Measured in J/kg.K

✓Latent Heat

- The amount of heat absorbed by a unit mass of a substance at a *change of state with no change in temperature*
- Measured in J/kg

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Physics of Heat

$$Q = c * m * \Delta T$$

What is 'Specific Heat'?

- ✓ The specific heat is the amount of heat per unit mass required to raise the temperature by one degree Celsius.
- ✓ The relationship does not apply if a phase change is encountered, because the heat added or removed during a phase change does not change the temperature.
- ✓ Water plays a very important role in temperature regulation. The specific heat per gram for water is much higher than that for a metal

Thermodynamics

… and why it matters

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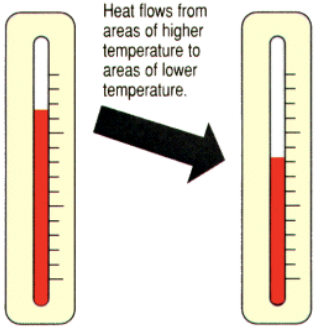
Thermodynamics

Thermodynamics

- ✓The 1st Law
  - ✓The principle of conservation of energy
  - ✓Energy cannot be created or destroyed, only moved.
- ✓The 2nd Law
  - ✓Heat (or energy) transfer can take place spontaneously in one direction
  - ✓Heat moves from hotter to cooler body



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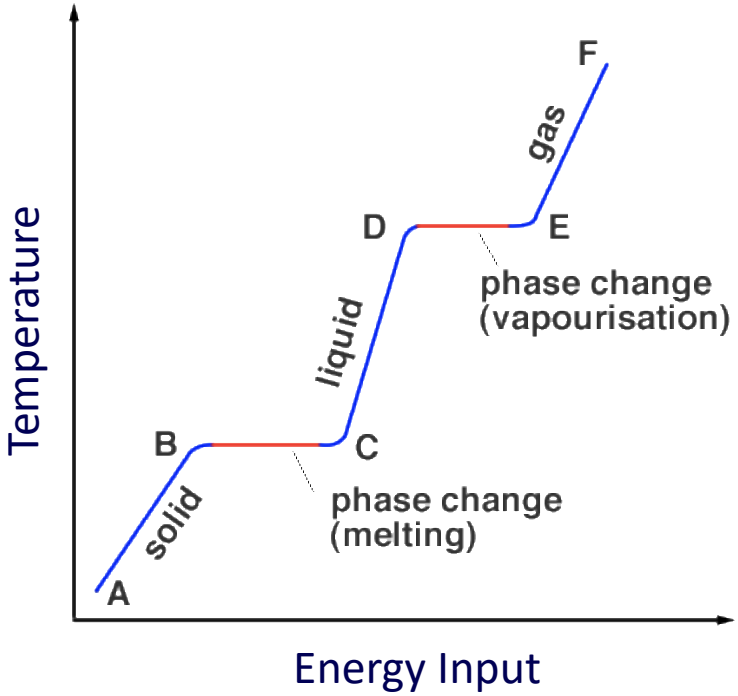
Physics of Heat

Heat, Temperature & Energy



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Thermodynamics

Thermal heat energy

- ✓Where does heat go?
  - ✓Heat flows from warmer to colder body/environments
- ✓Why does your cup of tea get colder?
  - ✓Because the environment around your tea is ***colder*** than your cup of tea

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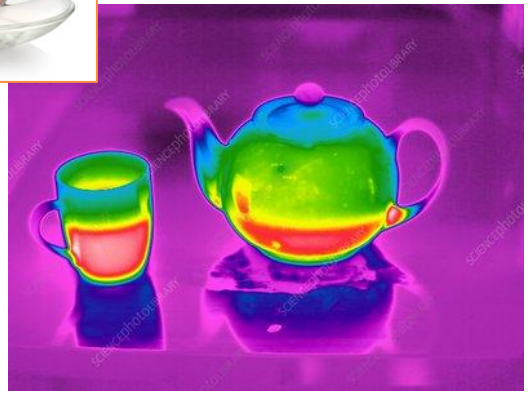
Thermodynamics

Thermal heat energy



- ✓What do you need to heat your cup of tea?
  - ✓Energy (e.g. fire, electricity)

- ✓What could you do to slow down cooling?
  - ✓Decrease heat loss (flux) by using insulation

We won't to avoid thermal bridges!

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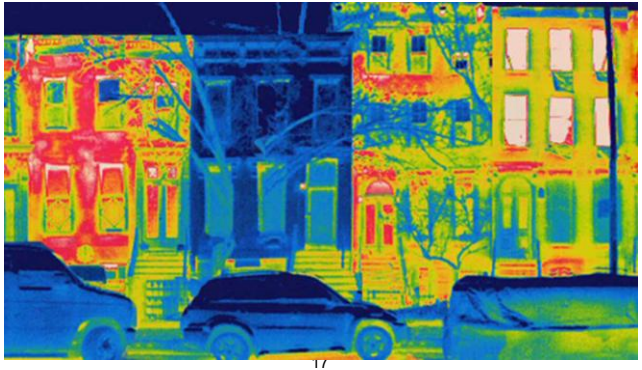
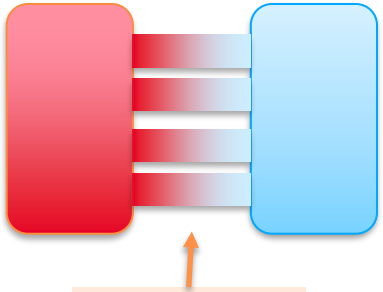
Thermodynamics

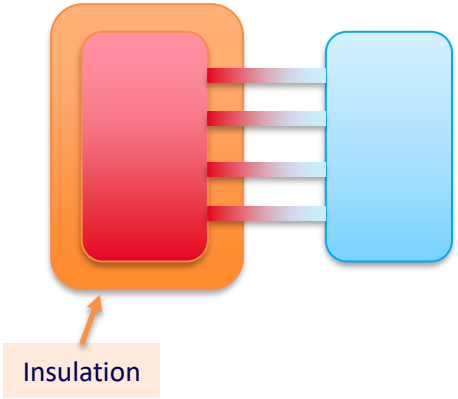
Thermal bridges

✓We must reduce thermal bridges in buildings for them to be energy efficient and sustainable

✓Infrared (IR) image of well-insulated house







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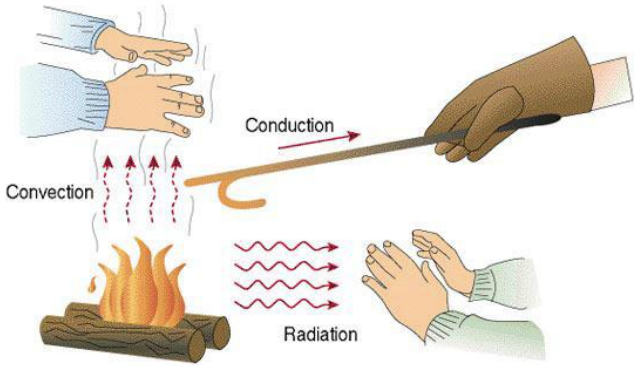
Thermodynamics

Heat Flow

✓Heat flow can take place in 3 forms:

- ✓ ***Conduction*** within a body or bodies in contact
- ✓ ***Convection*** within fluids (liquid or gas)
- ✓ ***Radiation*** from a body with a warmer surface to another which is cooler



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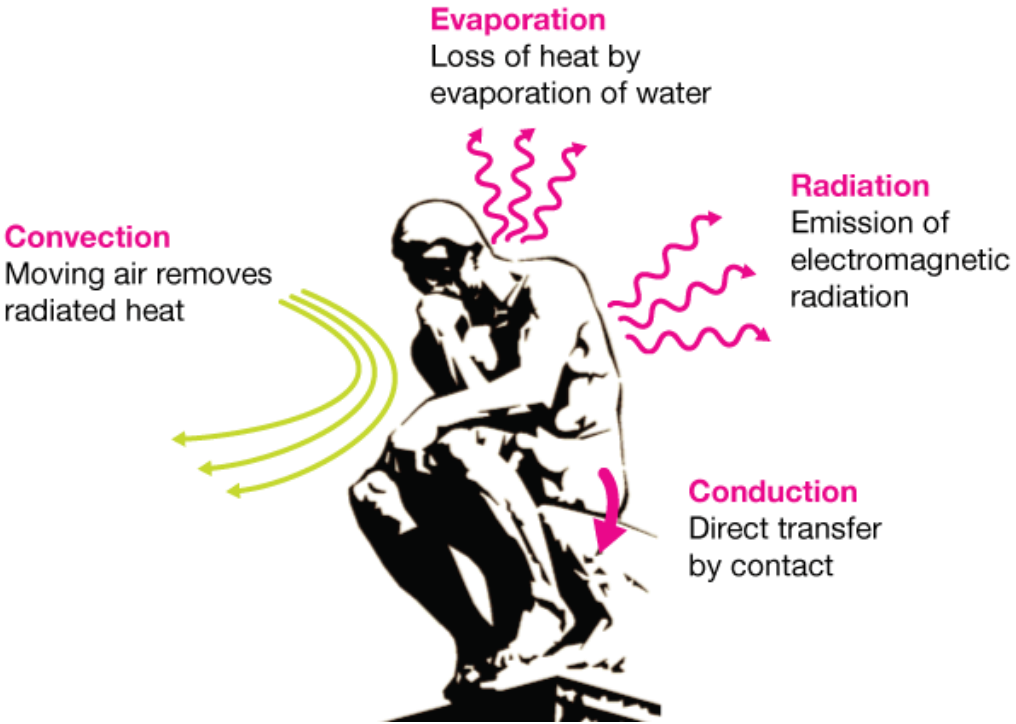
Thermodynamics

Heat Flow



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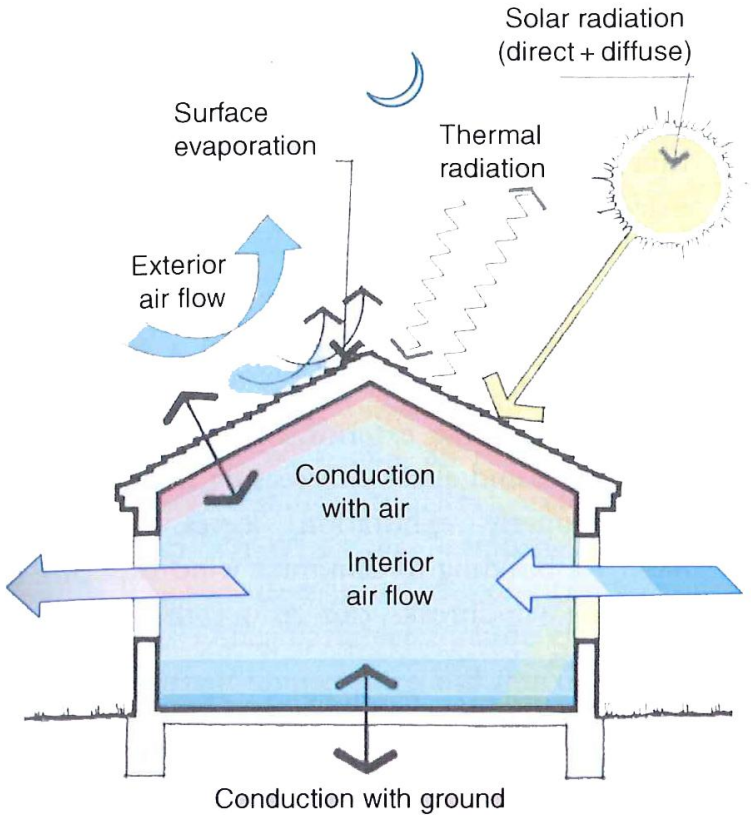
Thermodynamics

Heat Flow

Means of heat transport in ***buildings:***

- ✓ ***Conduction*** – from hotter objects to cooler objects by direct contact
- ✓ ***Convection*** – by flow of air between warmer objects and cooler objects
- ✓ ***Radiation*** – from hotter objects to cooler objects within the direct view of each other, including radiation from sun to earth
- ✓ ***Evaporation*** – change of phase from liquid to gaseous state



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Thermodynamics

Fourier's law

$$\frac{\Delta Q}{\Delta t} = \frac{\lambda}{d} * A * (T_1 - T_2)$$

Heat Flux

✓Fourier's law

✓Thermal conductivity (λ)
✓Thickness (d) is in metres (m)
✓Temperature difference across a boundary ($T_1 - T_2$) in Kelvin (K)
✓Time (t)

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Heat Flow

Conductivity

- ✓ Conductivity*(λ)*
  - A material property, regardless of its shape or size
  - Measured as the heat flow density (W/m^2) in a 1m thick body, with a one degree (1ºC) temperature difference
  - The rate of heat flow through a unit thickness of a homogeneous material
  - Measured in units of W/m.K

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Heat Flow

Conductivity *(λ)*

Example – Thermal properties of materials

		Conductivity (W/mK)	Density (kg/m^3)	Specific heat (J/kg K)
Wall materials	Brickwork, outer leaf	0.840	1700	800
	Concrete block, medium	0.510	1400	1000
	Timber, hardwood	0.150	680	1200
Insulating materials	Glass fibre quilt	0.040	12	840
	Polyurethane board	0.025	30	1400
	Rubber, sheet	0.160	930	2010
Miscellaneous	Air (25° C, 50% RH, still)	0.025	1.15	1063
	Water (still)	0.58	1000	4187
	Ice (-1° C)	2.200	918	

Source: Szokolay, 2008

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Heat Flow

Conductivity *(λ)*

Common materials used in Buildings	Thermal Conductivity, λ (W/mK)
Vacuum Insulated Panel	0.007
Polyurethane Foam Insulation (Spray Applied)	0.024
Extruded Polystyrene (XPS) with CFC and HCFC	0.029
Extruded Polystyrene (XPS) with CO2	0.034
Cellulose Fiber, Cotton Fiber (Dense pack)	0.035
Mineral Wool (Roxul…)	0.035
Expanded Polystyrene (EPS)	0.038
Glass Wool, Fibreglass Insulation	0.038
Wood-fibre insulation	0.04
Foam Glass	0.04
Cork insulation	0.045
Straw bale	0.066
OSB, Plywood, Softwood	0.13
Gypsum Board	0.25
Lightweight concrete	0.39
Concrete Reinforced	2.06
Steel - Stainless (Buffed)	17
Steel - Galvanized Sheet (0.14% C)	62
Aluminum (Anodized)	237

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Heat Flow

Conductivity *(λ)*

- *Some insulation has more 'embodied carbon than others*

Insulation Material	R-value R/inch	Density lb/ft³	Emb. E MJ/kg	Emb. Carbon $kgCO_2/kg$	Emb. Carbon $kgCO_2/ft^2 \cdot R$	Blowing Agent (GWP)	Bl. Agent kg/kg foam	Blowing Agent GWP/ bd-ft	Lifetime GWP/ $ft^2 \cdot R$
Cellulose (dense-pack)	3.7	3.0	2.1	0.106	0.0033	None	0	N/A	0.0033
Fiberglass batt	3.3	1.0	28	1.44	0.0165	None	0	N/A	0.0165
Rigid mineral wool	4.0	4.0	17	1.2	0.0455	None	0	N/A	0.0455
Polyisocyanurate	6.0	1.5	72	3.0	0.0284	Pentane (GWP=7)	0.05	0.02	0.0317
Spray polyurethane foam (SPF) – closed-cell (HFC-blown)	6.0	2.0	72	3.0	0.0379	HFC-245fa (GWP=1,030)	0.11	8.68	1.48
SPF – closed-cell (water-blown)	5.0	2.0	72	3.0	0.0455	Water (CO_2) (GWP=1)	0	0	0.0455
SPF – open-cell (water-blown)	3.7	0.5	72	3.0	0.0154	Water (CO_2) (GWP=1)	0	0	0.0154
Expanded polystyrene (EPS)	3.9	1.0	89	2.5	0.0307	Pentane (GWP=7)	0.06	0.02	0.036
Extruded polystyrene (XPS)	5.0	2.0	89	2.5	0.0379	HFC-134a[1] (GWP=1,430)	0.08	8.67	1.77

1. XPS manufacturers have not divulged their post-HCFC blowing agent, and MSDS data have not been updated. The blowing agent is assumed here to be HFC-134a.

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Heat Flow

Nominal vs Effective R-value

Nominal = R-value of insulation

Effective = R-value of overall assembly

(Includes effects of studs can be substantially lower.)

Studs make a difference!

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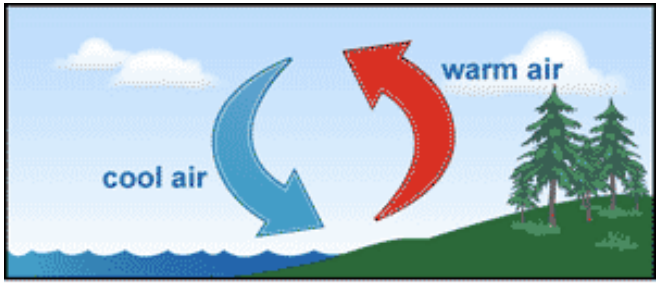
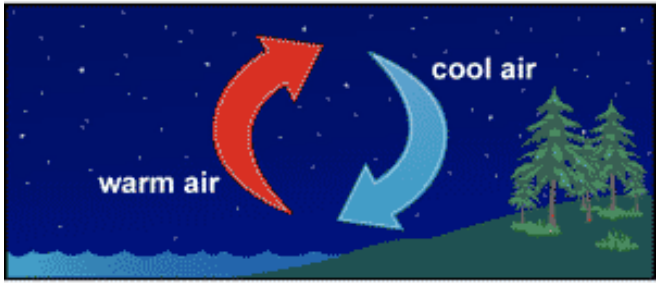
Heat Flow

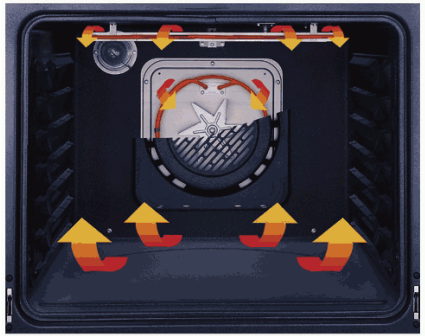
Convection

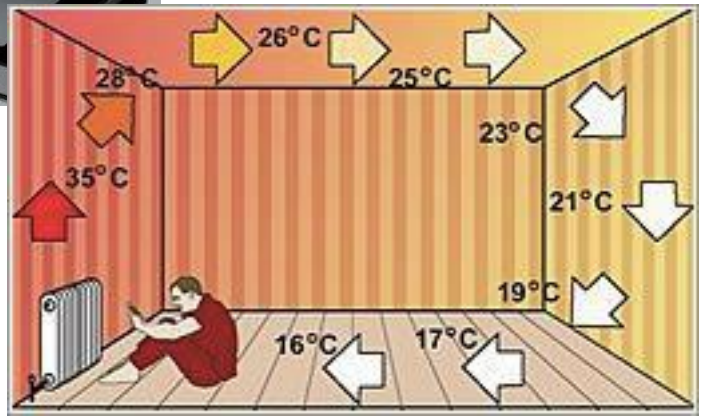


DAY TIME



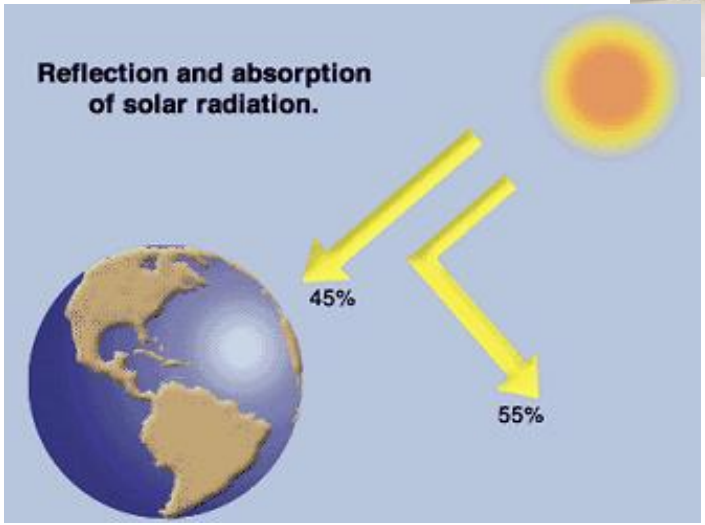
NIGHT TIME

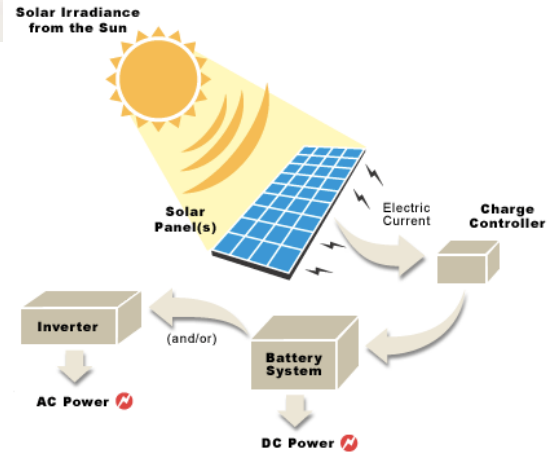




Heat Flow

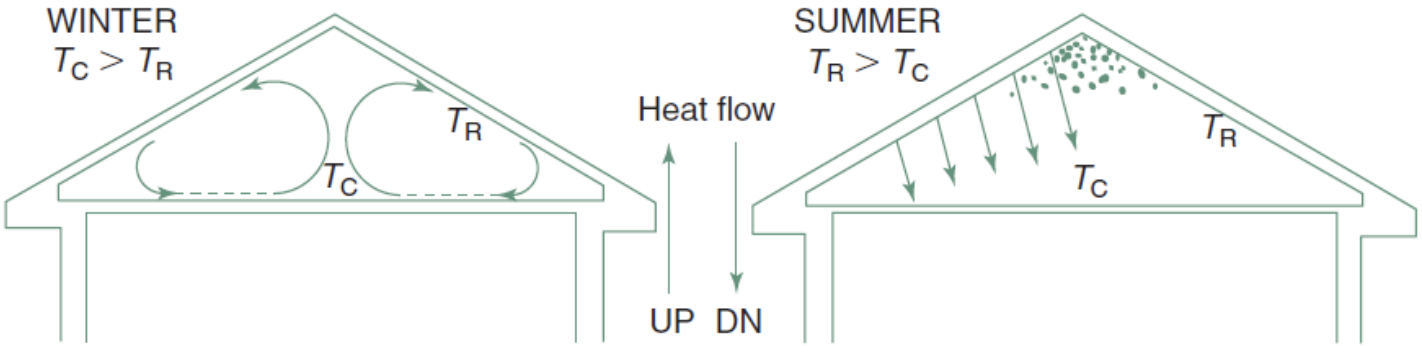
Radiation





Heat Flow

Application: Buildings



Air warmed by ceiling rises, it is cooled by the roof and drops back to the ceiling

Dominantly convection

Hot air accumulates and stays at the highest point

Dominantly radiation

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Thank you

Next time: Controlling heat transfer in buildings with Insulation

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